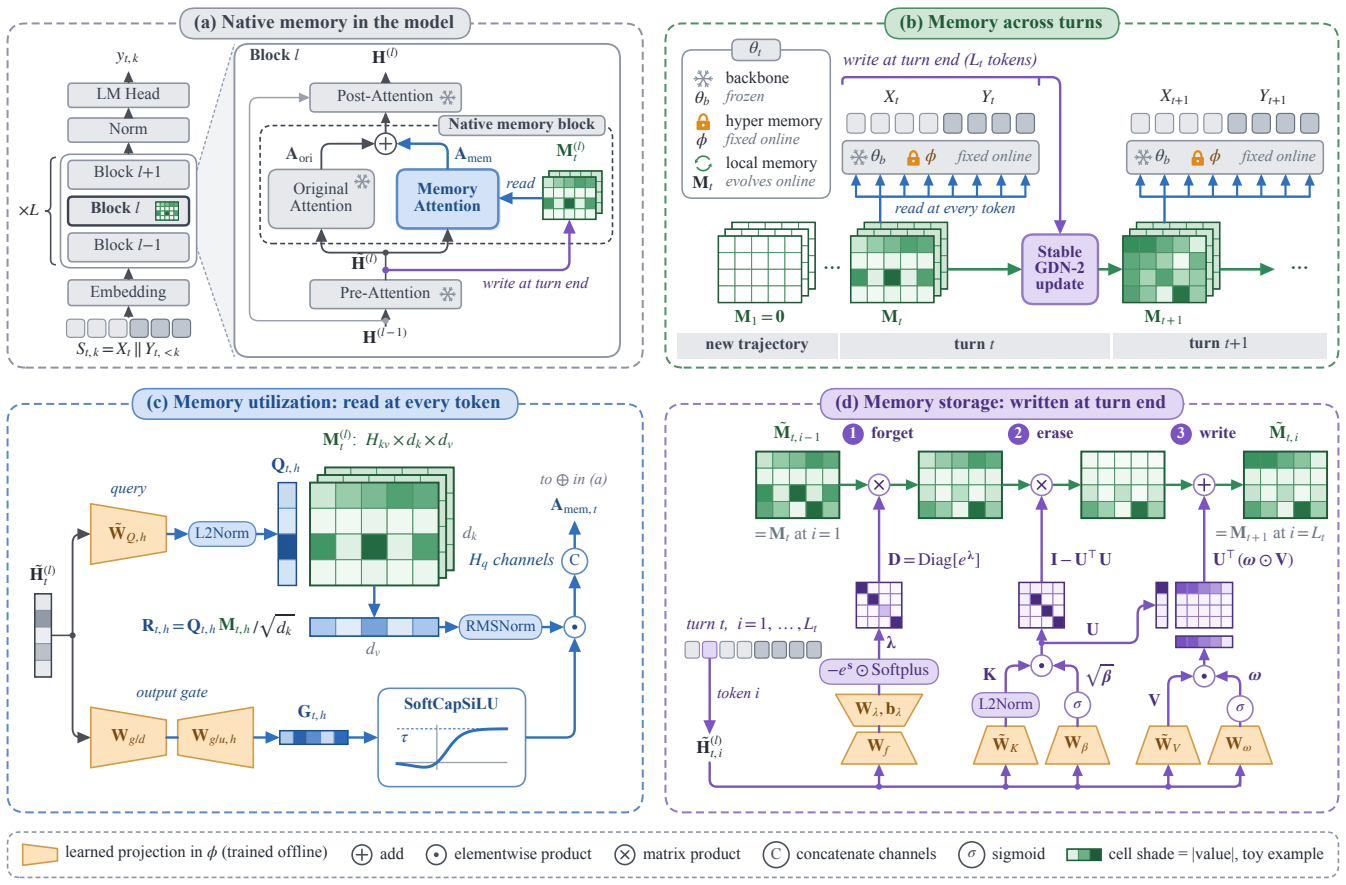

(a) Native memory in the model
$y_{t,k}$
LM Head
Norm
×L
Block l+1
Block l
Block l−1
Embedding
$S_{t,k} = X_t \,\|\, Y_{t,<k}$
Block l
$\mathbf{H}^{(l)}$
Post-Attention
Native memory block
$\mathbf{A}_{\text{ori}}$
$\mathbf{A}_{\text{mem}}$
$\mathbf{M}_t^{(l)}$
Original Attention
Memory Attention
read
$\mathbf{H}^{(l)}$
write at turn end
Pre-Attention
$\mathbf{H}^{(l-1)}$
(b) Memory across turns
θ_t
backbone
θ_b frozen
hyper memory
ϕ fixed online
local memory
$\mathbf{M}_t$ evolves online
write at turn end (L_t tokens)
X_t
Y_t
θ_b ϕ fixed online
read at every token
X_{t+1}
Y_{t+1}
θ_b ϕ fixed online
$\mathbf{M}_1 = \mathbf{0}$
$\mathbf{M}_t$
Stable GDN-2 update
$\mathbf{M}_{t+1}$
new trajectory
turn t
turn t+1
(c) Memory utilization: read at every token
$\mathbf{M}_t^{(l)}$: $H_{kv} \times d_k \times d_v$
$\mathbf{Q}_{t,h}$
query
$\tilde{\mathbf{W}}_{Q,h}$
L2Norm
d_k
H_q channels
to ⊕ in (a)
$\mathbf{A}_{\text{mem},t}$
C
$\tilde{\mathbf{H}}_t^{(l)}$
$\mathbf{R}_{t,h} = \mathbf{Q}_{t,h}\mathbf{M}_{t,h}/\sqrt{d_k}$
RMSNorm
d_v
output gate
$\mathbf{W}_{g/d}$
$\mathbf{W}_{g/u,h}$
$\mathbf{G}_{t,h}$
SoftCapSiLU
τ
(d) Memory storage: written at turn end
$\tilde{\mathbf{M}}_{t,i-1}$
1 forget
2 erase
3 write
$\tilde{\mathbf{M}}_{t,i}$
$= \mathbf{M}_t$ at $i = 1$
$\mathbf{D} = \mathrm{Diag}[e^{\lambda}]$
$\mathbf{I} - \mathbf{U}^\top\mathbf{U}$
$\mathbf{U}^\top(\boldsymbol{\omega} \odot \mathbf{V})$
$= \mathbf{M}_{t+1}$ at $i = L_t$
turn t, $i = 1, \ldots, L_t$
token i
$\tilde{\mathbf{H}}_{t,i}^{(l)}$
$\boldsymbol{\lambda}$
$-e^{s} \odot \mathrm{Softplus}$
$\mathbf{W}_\lambda, \mathbf{b}_\lambda$
$\mathbf{W}_f$
$\mathbf{U}$
$\mathbf{K}$
L2Norm
$\tilde{\mathbf{W}}_K$
$\sqrt{\boldsymbol{\beta}}$
σ
$\mathbf{W}_\beta$
$\mathbf{V}$
$\tilde{\mathbf{W}}_V$
$\boldsymbol{\omega}$
σ
$\mathbf{W}_\omega$
learned projection in ϕ (trained offline)
add
elementwise product
matrix product
concatenate channels
sigmoid
cell shade = |value|, toy example